

# An approach for maximizing the smallest eigenfrequency of structure vibration based on piecewise constant level set method



Zhengfang Zhang<sup>a</sup>, Weifeng Chen<sup>b,\*</sup>

<sup>a</sup> College of Science, Hangzhou Dianzi University, Hangzhou 310018, PR China

<sup>b</sup> School of Information, Zhejiang University of Finance and Economics, Hangzhou 310018, PR China

## ARTICLE INFO

### Article history:

Received 8 August 2017

Received in revised form 28 January 2018

Accepted 29 January 2018

Available online 2 February 2018

### Keywords:

Piecewise constant level set method

Elasticity system

Sensitivity analysis

Penalty gradient algorithm

The first variation

## ABSTRACT

Maximization of the smallest eigenfrequency of the linearized elasticity system with area constraint is investigated. The elasticity system is extended into a large background domain, but the void is vacuum and not filled with ersatz material. The piecewise constant level set (PCLS) method is applied to present two regions, the original material region and the void region. A quadratic PCLS function is proposed to represent the characteristic function. Consequently, the functional derivative of the smallest eigenfrequency with respect to PCLS function takes nonzero value in the original material region and zero in the void region. A penalty gradient algorithm is proposed, which initializes the whole background domain with the original material and decreases the area of original material region till the area constraint is satisfied. 2D and 3D numerical examples are presented, illustrating the validity of the proposed algorithm.

© 2018 Elsevier Inc. All rights reserved.

## 1. Introduction

Structural shape and topology optimization is a tool to achieve maximum efficiency in a structure design. Four major methods have been developed in the past decades [1]: homogenization method [2], evolutionary structural optimization method (ESO) [3], solid isotropic material with penalization (SIMP) method [4] and level set method (LSM) [5].

LSM, proposed by Osher and Sethian [6], is the latest method in the four major methods, and is widely applied as a powerful tool to deal with the structural optimization problems. It is first introduced by Sethian and Wiegmann to solve the boundary design of elastic structures [7]. Osher and Santosa investigate the maximization of the smallest eigenvalue and the gap of first two eigenvalues in the scalar eigenvalue partial differential equation [8]. Wang et al. first introduce the level set method to structural topology optimization [5]. Allaire et al. use level set method for two optimization problems [9], the minimization of compliance and the minimization of the least square error of the computed displacement and the target one. They extend it to deal with vibration and multiple loads structural optimization problems [10]. Gournay proposes an extension of the velocity of the underlying Hamilton–Jacobi equation to improve the rate of convergence of the level-set method [11]. Xia et al. [12] using level set method to deal with the maximizing the simple and repeated first eigenvalue of structure vibration. For the purpose of structure vibration reduction, the minimization of the frequency response at the specified points or surfaces on the structure with an excitation frequency or a frequency range are investigated by Shu et

\* Corresponding author.

E-mail address: cwf818@zufe.edu.cn (W. Chen).

al. [13]. Theillard et al. [14] present a second-order sharp numerical method for solving the linear elasticity equations on irregular domains. They also apply the numerical scheme to maximize the stiffness of a structure based on level set method. For the optimization of structures with multiple-type boundaries, a level set method is proposed to represent and propagate different segments of boundary of a geometric shape [15,16]. A novel stress-related objective function based on the global stress-deviation measure is proposed by Polajnar et al. [17].

Piecewise constant level set (PCLS) method is an alternative level set method, which is proposed by Christiansen and Tai [18]. There are two main improvements over the traditional level set method. First, the initialization of PCLS function is more general. It is unnecessary to dig holes when the PCLS function is initialized. In the process of iteration, holes can be created and eliminated automatically. Second, there is no re-initialization process for the PCLS method. Alternatively, a simple constraint is added, keeping the PCLS function to be piecewise constants.

The PCLS method is originally applied to image segmentation and leads to satisfactory segmentation results [19]. To solve the structural shape and topology optimization problem, PCLS method is introduced and implemented as well. Wei and Wang [20] first apply PCLS method to minimize the mean compliance of elastic structure based on the ersatz material method. For the same objective functional, the variational binary level set method is applied by Dai et al. [21]. PCLS method is extended into the boundary control of eigenvalue optimization problem by Zhang et al. [22] and elastic contact problems by Myslinski [23].

For the structural optimization problem, the LSM and PCLS method are based on the ersatz material approach [10,20]. That is, the original elasticity system is extended into a large background domain, filling the void with the weak materials. This approach provides a good approximation to the original structural optimization problem. The advantage is that the computational domain is fixed, which is the whole background domain. The mesh partitions and the dimensions of the corresponding discretization matrices are fixed. However, it incurs a problem. Because of the huge jump of the coefficients on the interface of the original material and the weak material, the solution of the extended elasticity system has a large numerical error on the interface.

To overcome the abovementioned difficulty, the artificial weak material is removed and the body-fitted mesh is used. Ha and Cho use re-meshing technique in the actual domain for the topology optimization of nonlinear hyperelastic structures based on the level set method [24]. Zhou et al. use adaptive mesh to solve the design problem of the dipole antenna and develop a level-set framework for electromagnetic topology optimization problems [25]. Xia et al. propose a novel global measure of von Mises stress [26]. They avoid the artificial weak material and modify a fixed background mesh to conduct the finite element analysis.

In this paper, we adopt the idea to set up a background domain, but keep the void empty and not use ersatz material to fill it. There are two advantages of this method. First, the background domain is fixed and the meshes are partitioned once at the beginning. We do not need to repartition the original material domain time to time in the process of iteration. Second, there is no approximation to the original structure optimization. Only the original material region is used in the computation of elasticity system, not the whole background domain. Then we use the PCLS to present two regions, the original material region and the void region. A quadratic function is constructed to represent the characteristic function of the original region. This quadratic function guarantees the functional derivative of the smallest eigenfrequency with respect to PCLS function takes nonzero value in the original material region and zero in the void region.

We focus on maximization of the smallest eigenfrequency, govern by eigenvalue problem for the linearized elasticity system. Using continuum sensitivity analysis, the functional derivative of the smallest eigenfrequency with respect to the PCLS function is obtained. The penalty method is used to deal with the area constraint. The additional constraint for PCLS function is fulfilled by a projection. The total variation regularization term is applied, to control the perimeter of the subdomain boundaries as well as the jumps of the PCLS function. This strategy effectively avoids the emergence of isolated region and the localized mode. For the same purpose, Li et al. investigate a mode recognition technique based on level set method to eliminate the localized eigenmodes [27]. In this paper, we propose a penalty gradient algorithm to update the PCLS function. When the smallest eigenfrequency becomes multiple, the strategy proposed by [28] is used to follow the correct eigenmode to avoid jumping back and forth among different eigenmodes. Three numerical experiments of 2D cantilevers are conducted in square domains and two numerical experiments of 3D cantilevers are conducted in a rectangular parallelepiped.

This paper is organized as follows. In Section 2, the maximization problem of smallest eigenfrequency of linearized elasticity system is described. In Section 3, PCLS method is introduced and the first variation of smallest eigenfrequency with respect to the PCLS is derived. A penalty gradient algorithm is proposed in Section 4. In Section 5, we present the numerical results of five cantilever problems in 2D and 3D domains. The last section gives some concluding remarks.

## 2. Problem statement

Consider the eigenvalue problem for the linearized elasticity system

$$-\nabla \cdot (\mathbb{A}e(u)) = \omega^2 \rho u, \quad \text{in } \Omega, \quad (1)$$

$$u = 0, \quad \text{on } \Gamma_D, \quad (2)$$

$$\mathbb{A}e(u)n = 0, \quad \text{on } \Gamma_N, \quad (3)$$

where  $\Omega$  is a smooth, bounded and connected subset of  $\mathbb{R}^d$  ( $d = 2$  or  $3$ ).  $\Gamma_D$  and  $\Gamma_N$  are the Dirichlet boundaries and Neumann boundaries respectively, which are two disjoint parts of the boundary of  $\Omega$ .  $n$  is the unit outward normal to the boundary.  $\mathbb{A}$  describes the Hooke's law. For any symmetric matrix  $\xi$ , it is defined by

$$\mathbb{A}\xi = \lambda(\text{Tr}\xi)\text{Id} + 2\mu\xi, \quad (4)$$

where  $\lambda$  and  $\mu$  are the Lamé moduli of the material,  $\text{Tr}\xi$  is the trace of  $\xi$  and  $\text{Id}$  is the identity matrix.  $\rho > 0$  is the density of the material.  $e(u)$  is the strain tensor, which is defined by  $e(u) = 1/2(\nabla u + (\nabla u)^\top)$ , where  $\top$  is the transpose of a matrix. We denote by  $\omega \geq 0$  the vibration frequency and by  $u$  the associated mode, i.e. the corresponding displacement field in  $\Omega$ . The equation system (1)–(3) admits a countable infinite solutions  $(\omega_k, u_k)_{k \geq 1}$  in  $\mathbb{R}^+ \times H^1(\Omega)^d$ , labeled by increasing order of the vibration frequency.

Using integration by parts for (1) with boundary conditions (2) and (3), we have the weak formulation

$$\int_{\Omega} \mathbb{A}e(u) : e(v) dx = \omega^2 \int_{\Omega} \rho u v dx, \quad \forall v \in U(\Omega), \quad (5)$$

where  $v$  is a test vector function and

$$U(\Omega) = \left\{ v \in H^1(\Omega)^d : \int_{\Omega} \rho |v|^2 dx = 1, v = 0 \text{ on } \Gamma_D \right\}. \quad (6)$$

The first (i.e. smallest) eigenfrequency is characterized as the minimum value of the Rayleigh quotient [29]

$$\omega_1^2(\Omega) = \min_{u \in U(\Omega), u \neq 0} \frac{\int_{\Omega} \mathbb{A}e(u) : e(u) dx}{\int_{\Omega} \rho |u|^2 dx} \quad (7)$$

$$= \frac{\int_{\Omega} \mathbb{A}e(u_1(\Omega)) : e(u_1(\Omega)) dx}{\int_{\Omega} \rho |u_1(\Omega)|^2 dx}, \quad (8)$$

where  $u_1(\Omega)$  is the first eigenmode associated with  $\omega_1^2(\Omega)$ . The optimization problem in this paper is to find the optimal shape of  $\Omega$  with certain area constraint to maximize the first eigenfrequency  $\omega_1^2(\Omega)$ . We introduce a background domain  $D$  (a bounded open set of  $\mathbb{R}^d$ ) which contains all admissible  $\Omega$ . The set of admissible  $\Omega$  is defined by

$$\Omega_{Ad} = \left\{ \Omega \subset D : |\Omega| = V \right\}, \quad (9)$$

where  $V$  is a given area. Therefore, the optimization problem is defined by

$$\inf_{\Omega \in \Omega_{Ad}} J(\Omega), \quad (10)$$

where

$$J(\Omega) = -\omega_1^2(\Omega). \quad (11)$$

Considering the introduced background domain  $D$ , we extend the elasticity equation system into  $D$  by

$$-\nabla \cdot (\chi_{\Omega} \mathbb{A}e(u)) = \omega^2 \chi_{\Omega} \rho u, \quad \text{in } D, \quad (12)$$

$$u = 0, \quad \text{on } \partial D_D, \quad (13)$$

$$\chi_{\Omega} \mathbb{A}e(u)n = 0, \quad \text{on } \partial D_N, \quad (14)$$

$$\chi_{\Omega} \mathbb{A}e(u)n = 0, \quad \text{on } \Gamma_N, \quad (15)$$

where  $\chi_{\Omega}$  is the characteristic function of  $\Omega$ . The boundary of  $D$  is decomposed into two parts:  $\partial D_D$  corresponds to Dirichlet boundary conditions and  $\partial D_N$  to homogeneous Neumann boundary conditions. Recall the boundary conditions (2) and (3), admissible shapes  $\Omega$  are further constrained to satisfy

$$\Gamma_D \subset \partial D_D, \quad \Gamma_N \cap \partial D_D = \emptyset. \quad (16)$$

That is, the boundary  $\Gamma_D$ , with zero displacement, must be a subset of the fixed boundary  $\partial D_D$ , while the homogeneous Neumann boundary  $\Gamma_N$  can be anywhere but on  $\partial D_D$ .

Using integration by parts for (12) with boundary conditions (13)–(14) and the interface condition (15), we have the weak formulation

$$\int_D \chi_{\Omega} \mathbb{A}e(u) : e(v) dx = \omega^2 \int_D \chi_{\Omega} \rho u v dx, \quad \forall v \in U^*(D), \quad (17)$$

where  $v$  is a test vector function and

$$U^*(D) = \left\{ v \in H^1(D)^d : v|_{\Omega} \in U(\Omega), \chi_{\Omega} \mathbb{A}e(u)n = 0, \text{ on } \Gamma_N \text{ and } u = 0, \text{ on } \partial D_D \right\}. \quad (18)$$

According to the Rayleigh quotient, the first eigenfrequency of (17) is obtained by

$$\omega_{1,D}^2(\Omega) = \min_{u \in U^*(D), u \neq 0} \frac{\int_D \chi_{\Omega} \mathbb{A}e(u) : e(u) dx}{\int_D \chi_{\Omega} \rho |u|^2 dx} \quad (19)$$

$$= \frac{\int_D \chi_{\Omega} \mathbb{A}e(u_{1,D}(\Omega)) : e(u_{1,D}(\Omega)) dx}{\int_D \chi_{\Omega} \rho |u_{1,D}(\Omega)|^2 dx} \quad (20)$$

where  $u_{1,D}(\Omega)$  is the first eigenmode associated with  $\omega_{1,D}^2(\Omega)$ . We have the following proposition.

**Proposition 2.1.** Let  $\omega_1^2(\Omega)$  be the first eigenfrequency of (5) and  $\omega_{1,D}^2(\Omega)$  be the first eigenfrequency of (17). Then  $\omega_1^2(\Omega) = \omega_{1,D}^2(\Omega)$ .

**Proof.** By the definition of (7) and (19), we have

$$\begin{aligned} \omega_{1,D}^2(\Omega) &= \min_{u \in U^*(D), u \neq 0} \frac{\int_D \chi_{\Omega} \mathbb{A}e(u) : e(u) dx}{\int_D \chi_{\Omega} \rho |u|^2 dx} \\ &= \min_{u|_{\Omega} \in U(\Omega), u \neq 0} \frac{\int_D \chi_{\Omega} \mathbb{A}e(u) : e(u) dx}{\int_D \chi_{\Omega} \rho |u|^2 dx} \\ &= \min_{u \in U(\Omega), u \neq 0} \frac{\int_{\Omega} \mathbb{A}e(u) : e(u) dx}{\int_{\Omega} \rho |u|^2 dx} \\ &= \omega_1^2(\Omega) \quad \square \end{aligned}$$

Therefore, using the extended elasticity system (12)–(15), the original optimization problem (10) is equivalent to

$$\inf_{\Omega \in \Omega_{Ad}} J(\Omega) = \inf_{\Omega \in \Omega_{Ad}} -\omega_1^2(\Omega) = \inf_{\Omega \in \Omega_{Ad}} -\omega_{1,D}^2(\Omega). \quad (21)$$

### 3. Piecewise constant level set method and sensitivity analysis

#### 3.1. Piecewise constant level set method

To deal with the extended elasticity system (12)–(15), we introduce a PCLS function

$$\phi(x) = \begin{cases} 1, & x \in \Omega, \\ 2, & x \in D \setminus \Omega. \end{cases} \quad (22)$$

A quadratic function is defined to represent the characteristic function. That is,

$$\chi_{\Omega} = (\phi - 2)^2. \quad (23)$$

Thus, the linearized elasticity system (12)–(15) is modified to be dependant on the PCLS function  $\phi$ . That is,

$$-\nabla \cdot ((\phi - 2)^2 \mathbb{A}e(u(\phi))) = \omega^2(\phi)(\phi - 2)^2 \rho u(\phi), \quad \text{in } D, \quad (24)$$

$$u(\phi) = 0, \quad \text{on } \partial D_D, \quad (25)$$

$$(\phi - 2)^2 \mathbb{A}e(u(\phi))n = 0, \quad \text{on } \partial D_N, \quad (26)$$

$$(\phi - 2)^2 \mathbb{A}e(u(\psi))n = 0, \quad \text{on } \Gamma_N, \quad (27)$$

The weak formulation of the linearized elasticity system (12)–(15) is obtained by

$$\int_D (\phi - 2)^2 \mathbb{A}e(u(\phi)) : e(v) dx = \omega^2(\phi) \int_D (\phi - 2)^2 \rho u(\phi) v dx, \quad \forall v \in U^*(D). \quad (28)$$

The first eigenfrequency of (28) is obtained by Rayleigh quotient that

$$\omega_1^2(\phi) = \min_{u \in U^*(D), u \neq 0} \frac{\int_D (\phi - 2)^2 \mathbb{A}e(u) : e(u) dx}{\int_D (\phi - 2)^2 \rho |u|^2 dx} \quad (29)$$

$$= \frac{\int_D (\phi - 2)^2 \mathbb{A}e(u_1(\phi)) : e(u_1(\phi)) dx}{\int_D (\phi - 2)^2 \rho |u_1(\phi)|^2 dx}, \quad (30)$$

where  $u_1(\phi)$  is the first eigenmode associated with  $\omega_1^2(\phi)$ . Using the PCLS function, the area of  $\Omega$  is presented by

$$|\Omega| = \int_D (\phi - 2)^2 dx. \quad (31)$$

A new constraint for  $\phi$  is added that

$$K(\phi) = (\phi - 1)(\phi - 2) = 0, \quad (32)$$

since the PCLS function needs to keep one of the two constants exactly. Due to the utility of the PCLS method, the optimization problem (21), which depends on  $\Omega$ , is converted into depending on PCLS function  $\phi$ . We summarize the constraint optimization problem as follows:

$$\inf_{\phi} J(\phi) = \inf_{\phi} -\omega_1^2(\phi) \quad (33)$$

$$\text{s.t. } \omega_1^2(\phi) = \frac{\int_D (\phi - 2)^2 \mathbb{A}e(u_1(\phi)) : e(u_1(\phi)) dx}{\int_D (\phi - 2)^2 \rho |u_1(\phi)|^2 dx}, \quad (34)$$

$$G(\phi) = \int_D (\phi - 2)^2 dx - V = 0, \quad (35)$$

$$K(\phi) = 0. \quad (36)$$

### 3.2. Sensitivity analysis

We attempt to use gradient-type algorithm to solve the optimization problems (33)–(36). Therefore, the gradient of the smallest eigenfrequency (34) with respect to PCLS function  $\phi$  is needed.

Taking the first variation of the smallest eigenfrequency  $\omega_1^2(\phi)$  in (34) with respect to  $\phi$  in the direction  $h$ , we obtain

$$\begin{aligned} \omega_1^{2'} h = & \left( \int_D ((2(\phi - 2)h \mathbb{A}e(u_1)) : e(u_1) + (\phi - 2)^2 \mathbb{A}e(u_1' h) : e(u_1) \right. \\ & + (\phi - 2)^2 \mathbb{A}e(u_1) : e(u_1' h)) dx \int_D (\phi - 2)^2 \rho |u_1|^2 dx \\ & - \int_D (\phi - 2)^2 \mathbb{A}e(u_1) : e(u_1) dx \left( \int_D 2(\phi - 2)h \rho |u_1|^2 dx \right. \\ & \left. \left. + 2 \int_D (\phi - 2)^2 \rho u_1 u_1' h dx \right) \right) / \left( \int_D (\phi - 2)^2 \rho |u_1|^2 dx \right)^2. \end{aligned} \quad (37)$$

Here and below, we omit the variable  $\phi$  for notation simplification.

It is easy to verify that

$$\mathbb{A}e(u' h) : e(u) = \mathbb{A}e(u) : e(u' h). \quad (38)$$

Hence, with  $u = u_1$  in (38), and  $v = u_1' h$  and  $v = u_1$  in (28), respectively, we obtain that

$$\begin{aligned} \omega_1^{2'} h = & \frac{1}{(\int_D (\phi - 2)^2 \rho |u_1|^2 dx)^2} \left( \int_D ((2(\phi - 2)h \mathbb{A}e(u_1)) : e(u_1) \right. \\ & \left. + 2(\phi - 2)^2 \mathbb{A}e(u_1) : e(u_1' h)) dx \int_D (\phi - 2)^2 \rho |u_1|^2 dx \right. \end{aligned}$$

$$\begin{aligned}
& - \int_D (\phi - 2)^2 \mathbb{A}e(u_1) : e(u_1) dx \left( \int_D 2(\phi - 2)h\rho|u_1|^2 dx \right. \\
& \left. + 2 \int_D (\phi - 2)^2 \rho u_1 u_1' h dx \right) \\
& = \frac{1}{(\int_D (\phi - 2)^2 \rho|u_1|^2 dx)^2} \left( \int_D ((2(\phi - 2)h\mathbb{A}e(u_1)) : e(u_1)) \right. \\
& \quad \left. + 2\omega_1^2 (\phi - 2)^2 \rho u_1 u_1' h dx \int_D (\phi - 2)^2 \rho|u_1|^2 dx \right. \\
& \quad \left. - \omega_1^2 \int_D (\phi - 2)^2 \rho|u_1|^2 dx \left( \int_D 2(\phi - 2)h\rho|u_1|^2 dx \right. \right. \\
& \quad \left. \left. + 2 \int_D (\phi - 2)^2 \rho u_1 u_1' h dx \right) \right) \\
& = 2 \frac{\int_D ((\phi - 2)h\mathbb{A}e(u_1) : e(u_1)) dx - \omega_1^2 \int_D (\phi - 2)h\rho|u_1|^2 dx}{\int_D (\phi - 2)^2 \rho|u_1|^2 dx} \\
& = 2 \frac{\int_D (\phi - 2)(\mathbb{A}e(u_1) : e(u_1) - \omega_1^2 \rho|u_1|^2) h dx}{\int_D (\phi - 2)^2 \rho|u_1|^2 dx} \tag{39}
\end{aligned}$$

The functional derivative of  $\omega_1^2$  with respect to  $\phi$  is evaluated by

$$D\omega_1^2 = 2 \frac{(\phi - 2)(\mathbb{A}e(u_1) : e(u_1) - \omega_1^2 \rho|u_1|^2)}{\int_D (\phi - 2)^2 \rho|u_1|^2 dx}. \tag{40}$$

#### 4. Algorithm

In this section, we propose a new algorithm based on PCLS to deal with the constraint eigenfrequency maximization problem (33)–(36). Similar to ersatz material method, we introduce a background domain  $D$ . The original elasticity system is extended into whole background domain. We partition the background domain once at the beginning. In the process of iteration, the meshes of the background are fixed. We pick out the elements of original material region, which is represented by  $\phi = 1$ , from the ones of the background domain, to assemble the stiffness and mass matrices. That is, our computation is focused on the original material region, not the whole background. The approximations and errors caused by the ersatz method are eliminated.

For the area constraint (35), we use the penalty method. That is, a penalty term related to the area constraint is added to the objective function. To overcome the ‘ill-posedness’, we use total variation regularization, which is used to control the perimeter of the subdomain boundaries as well as the jumps of the PCLS function. In the numerical experiments, this term helps to avoid the emergence of the isolated region and localized eigenmodes. The final penalty objective function is defined by

$$Q(\phi; \alpha, \beta) = -\omega_1^2(\phi) + \frac{1}{2\alpha} \left( \int_D (\phi - 2)^2 dx - V \right)^2 + \beta \int_D |\nabla \phi| dx, \tag{41}$$

where  $\alpha$  is the penalty coefficient and  $\beta$  is the regularity coefficient. For the constraint of  $\phi$  in (36), we use a projection to pull  $\phi$  back to one of the two constants. That is,

$$P_{\{1,2\}}\phi = \begin{cases} 1, & \phi < 1.5, \\ 2, & \phi \geq 1.5. \end{cases} \tag{42}$$

Based on the first variation of  $\omega_1^2$  in (39), the first variation of  $Q(\phi; \alpha, \beta)$  is obtained by

$$Q'(\phi; \alpha, \beta)h = -\omega_1^{2'}h + \frac{2}{\alpha} \left( \int_D (\phi - 2)^2 dx - V \right) \left( \int_D (\phi - 2)h dx \right) - \beta \int_D \nabla \cdot \left( \frac{\nabla \phi}{|\nabla \phi|} \right) h dx \tag{43}$$

Thus, we obtain the functional derivative of  $Q(\phi; \alpha, \beta)$  with respect to  $\phi$  by

$$DQ(\phi; \alpha, \beta) = -2 \frac{(\phi - 2)(\mathbb{A}e(u_1) : e(u_1) - \omega_1^2 \rho |u_1|^2)}{\int_D (\phi - 2)^2 \rho |u_1|^2 dx} + \frac{2}{\alpha} \left( \int_D (\phi - 2)^2 dx - V \right) (\phi - 2) - \beta \nabla \cdot \left( \frac{\nabla \phi}{|\nabla \phi|} \right). \quad (44)$$

Notice that the first two terms in functional derivative of  $Q(\phi; \alpha, \beta)$  (44) take nonzero value in the original material domain where  $\phi = 1$ , and take zero in the void region where  $\phi = 2$ . If we initialize the PCLS function  $\phi = 2$  in  $D$ , the functional derivative of  $Q(\phi; \alpha, \beta)$  with respect to  $\phi$  is zero. The iteration procedure could not be moved on. It is possible to change the value of the PCLS function from 1 to 2, but impossible to change the value of the PCLS function from 2 to 1. Thus, we initialize the whole background domain with the original materials, that is,  $\phi = 1$  in  $D$ . The region of the original material is shrinking till the area constraint is satisfied.

We use the steepest descent method to update  $\phi$ . That is,

$$d_k = -DQ(\phi^{(k)}; \alpha_k, \beta)$$

and in the  $k$ -th iteration,  $\phi$  is updated by

$$\phi^{(k+1)} = \phi^{(k)} + \tau^{(k)} d_k, \quad (45)$$

where

$$\tau^{(k)} = \arg \min_{\tau} Q(\phi^{(k)} + \tau d_k; \alpha, \beta).$$

In our algorithm, we set by

$$\tau^{(k)} = \zeta h / \max_{x \in D} |DQ(\phi^{(k)}, \alpha_k, \beta)|, \quad \zeta \in (0, 1), \quad (46)$$

where  $h$  is the size of domain meshes. To balance the value of  $\omega_1^2$  and the penalty term, the penalty coefficient  $\alpha_k$  is set by

$$\frac{1}{\alpha_k} = \max(\min(\frac{2\omega_1^2 t}{(\int_D (\phi - 2)^2 dx - V)^2}, \alpha_{\max}), \alpha_{\min}), \quad t \in (0, t_{\max}],$$

where  $t_{\max}$  is a given number. In our numerical experiment, we set  $t_{\max} \leq 10$ .  $\alpha_{\min}$  and  $\alpha_{\max}$  are two given numbers to control the parameter of  $\alpha_k$  not to be too small or too large.

As is well known, a symmetric structure incurs the multiplicity of eigenfrequencies. When the first eigenfrequency becomes multiple, the functional derivative of  $Q(\phi; \alpha, \beta)$  is not well-defined. It is necessary to specify which eigenmode is supposed to be used in (44). In our algorithm, we pick the closest one to the eigenmode  $u_1^{(k-1)}$  out of all eigenmodes whose associated eigenfrequencies are close to the current eigenvalue  $\omega_1^{2(k)}$  [28]. That is,

$$u_1^{(k)} = \arg \min_{u \in B_\delta} \| \pm u - u_1^{(k-1)} \|, \quad (47)$$

where, for a given  $\delta > 0$ , we set by

$$B_\delta = \left\{ u \in U^*(D) : u \text{ is an eigenmode associated with } \omega_j^{2(k)}, \text{ where } j \in J = \{j : |\omega_j^{2(k)} - \omega_1^{2(k)}| < \delta\} \right\}. \quad (48)$$

In our numerical experiments, this strategy is effective to track the correct eigenmode to calculate the functional derivative of  $Q(\phi; \alpha, \beta)$ .

Now, we summarize our new penalty gradient algorithm based on PCLS as follows:

**Algorithm 1.** Initialize  $\phi^{(0)} = 1$  in the whole background domain  $D$ . The constants  $\lambda$ ,  $\mu$ ,  $\rho$ ,  $\alpha_{\min}$ ,  $\alpha_{\max}$ ,  $\beta$ ,  $t$ ,  $\zeta$  and  $\delta$  are given.

Do for  $k = 0, 1, 2, \dots$

STEP 1. Compute the smallest eigenfrequency  $\omega_1^{2(k)}$  and the corresponding  $u_1^{(k)}$  by (1)–(3) using finite element method, where the  $\Omega^{(k)}$  is picked out by PCLS function  $\phi^{(k)} = 1$ . If  $\omega_1^{2(k)}$  ( $k > 0$ ) is multiple, select  $u_1^{(k)}$  by (47)–(48).

STEP 2. Compute  $DQ(\phi^{(k)}, \alpha_k, \beta)$  by (44) with  $\omega_1^{2(k)}$ ,  $u_1^{(k)}$  and  $\phi^{(k)}$ . Set the steepest decent direction by

$$d_k = -DQ(\phi^{(k)}, \alpha_k, \beta).$$

STEP 3. Update  $\phi^{(k+1)}$  by the inner iterations.

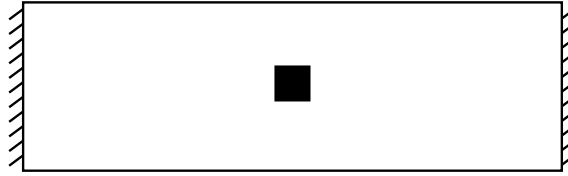


Fig. 1. Boundary condition for a 2D double cantilever. The black region is heavier and not subject to optimization.

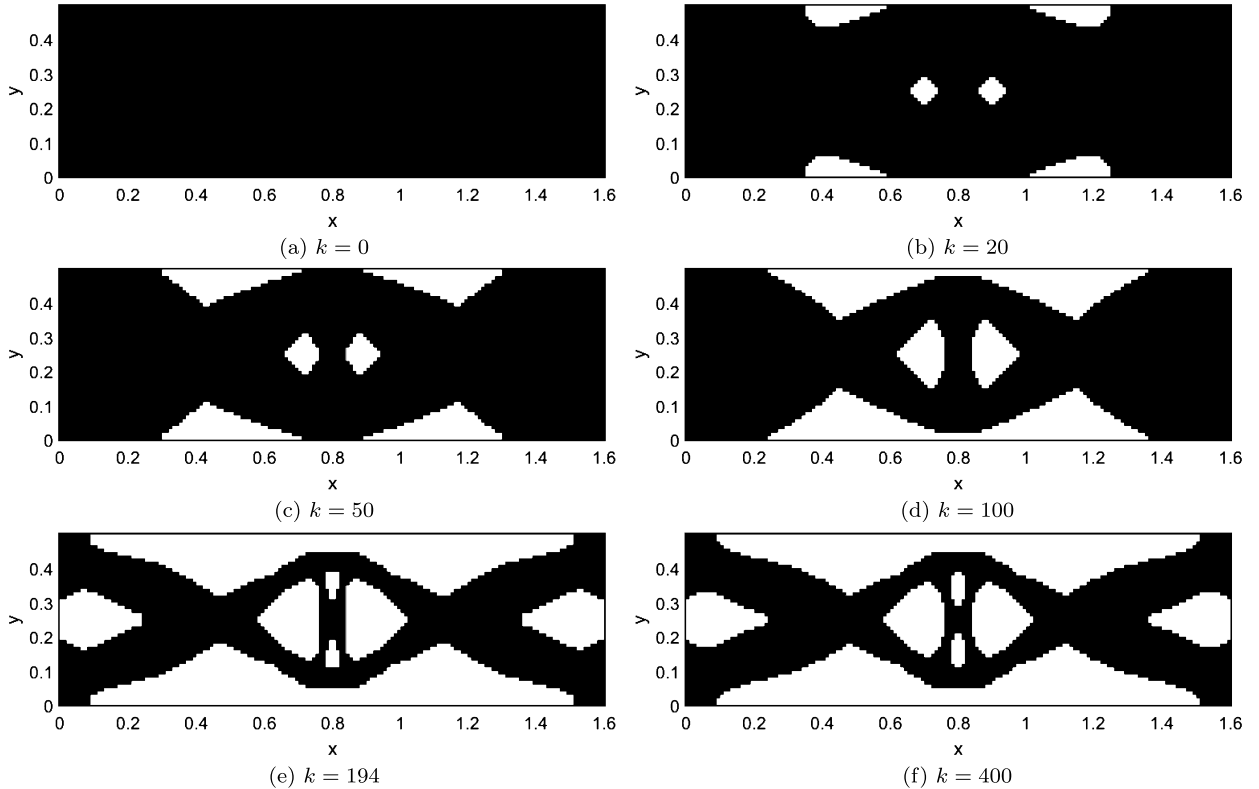


Fig. 2. The evolution of the double cantilever structure.

Do for  $p = 0, 1, 2, \dots$

STEP 3.1 Compute

$$\phi^* = \phi^{(k)} + \tau^{(k)} d_k,$$

where  $\tau^{(k)}$  is chosen by (46).

STEP 3.2 Project  $\phi^*$  using the projection defined in (42)

$$\phi^{**} = P_{\{1,2\}} \phi^*.$$

STEP 3.3 Check the stop criterion. If  $\phi^{**} \neq P_{\{1,2\}} \phi^{(k)}$ , we set  $\phi^{(k+1)} = \phi^{**}$  and go to STEP 5. Otherwise,  $\phi^{(k)} = \phi^*$ ,  $p = p + 1$  and go to STEP 3.1.

STEP 4. Check the convergence. If it is convergent, stop. Otherwise,  $k = k + 1$  and go to STEP 1.

**Remark 1.** Two stopping criteria are chosen to check the convergence. One is to set a maximum iteration step number. The other is to check if the area constraint is satisfied. In our numerical experiments, we observe that the extra iterations when the area constraint is satisfied alternate the structure little. Therefore, the convergence criterion in STEP 4 is that the iteration is stopped if the area constraint is satisfied.

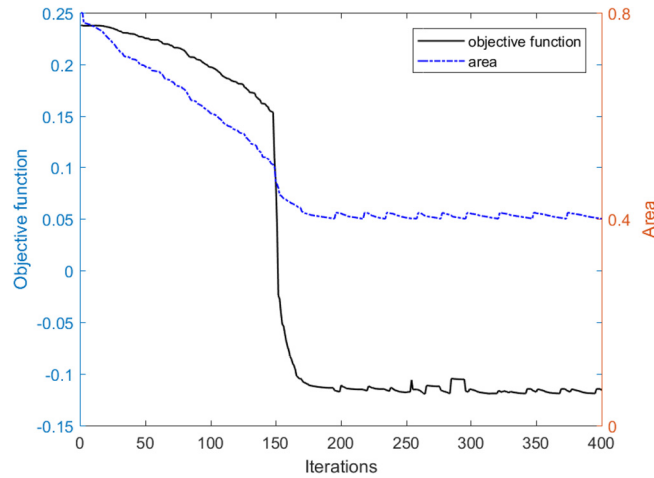


Fig. 3. The iteration histories of the objective function and the volume of the original material region.

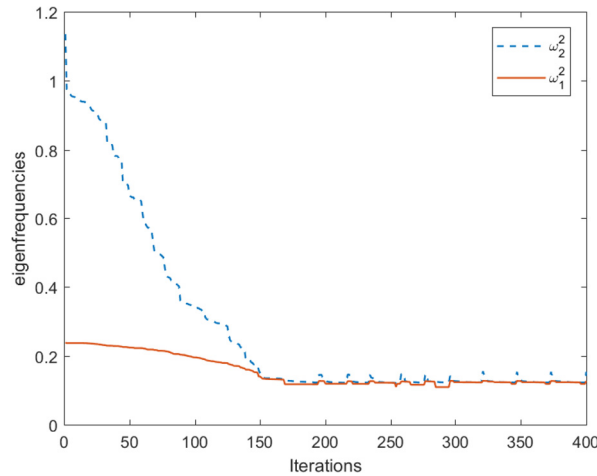


Fig. 4. The iteration histories of the first two eigenfrequencies.

## 5. Numerical results

In this section, we present some numerical results on experiments of proposed penalty gradient algorithm based on PCLS for the smallest eigenfrequency maximization problem (33)–(36) in 2D and 3D domains. To solve the elasticity equation (1)–(3), we use finite element method. In 2D cases, the regular rectangular meshes and piecewise bilinear bases are used. In 3D cases, the regular cubic meshes and piecewise trilinear bases are used. To assemble the stiffness and mass matrices, only the elements of original material region are picked out from the background elements. In our computation, the interface does not traverse the element. That is, the interface is just a part of the edges or faces of the elements. All test cases have the following data, unless otherwise specified. The Young modulus  $E$  of original material is normalized to 1, the Poisson ratio  $\nu$  is fixed to 0.3 and the density of original material is set to 1. The fixed area is set by  $V = \frac{|D|}{2}$ . We solve the elasticity equation system (1)–(3) and update the PCLS function by (45), using MATLAB codes.

### 5.1. 2D cantilever

For the first experiment, we consider a double cantilever problem. The domain  $D$  is of size  $1.6 \times 0.5$ . It is partitioned into  $160 \times 50$  rectangular elements. The left and right sides of the rectangular domain are set to zero Dirichlet boundary conditions, and the rest boundaries are set to zero Neumann boundary conditions. A small square region is set at the center of the rectangular domain (see Fig. 1), which is 100 times heavier and is not subject to optimization.

We set  $\alpha_{\min} = 1E-6$ ,  $\alpha_{\max} = 100$ ,  $\beta = 5E-4$ ,  $t = 2$ ,  $\zeta = 0.5$  and  $\delta = 1E-10$ . The initial PCLS function is set to  $\phi^{(0)}(x) = 1$  and is black colored in the whole domain as illustrated in Fig. 2 (a). The evolution profiles of PCLS function are illustrated in Fig. 2 (b)–(f), after  $k = 20, 50, 100, 194$  and 400 iterations, respectively. The black region indicates the original material, while the white region indicates the void. After 194 iterations, the area constraint is satisfied (see Fig. 2 (e)). The first

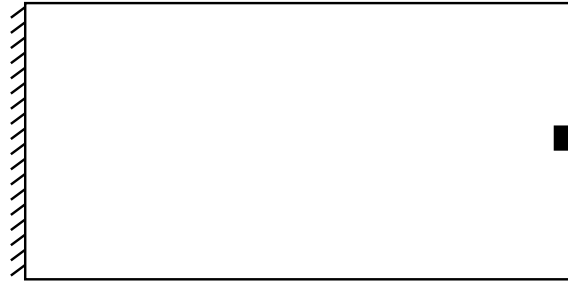


Fig. 5. Boundary condition for a 2D cantilever. The black region is heavier and not subject to optimization.

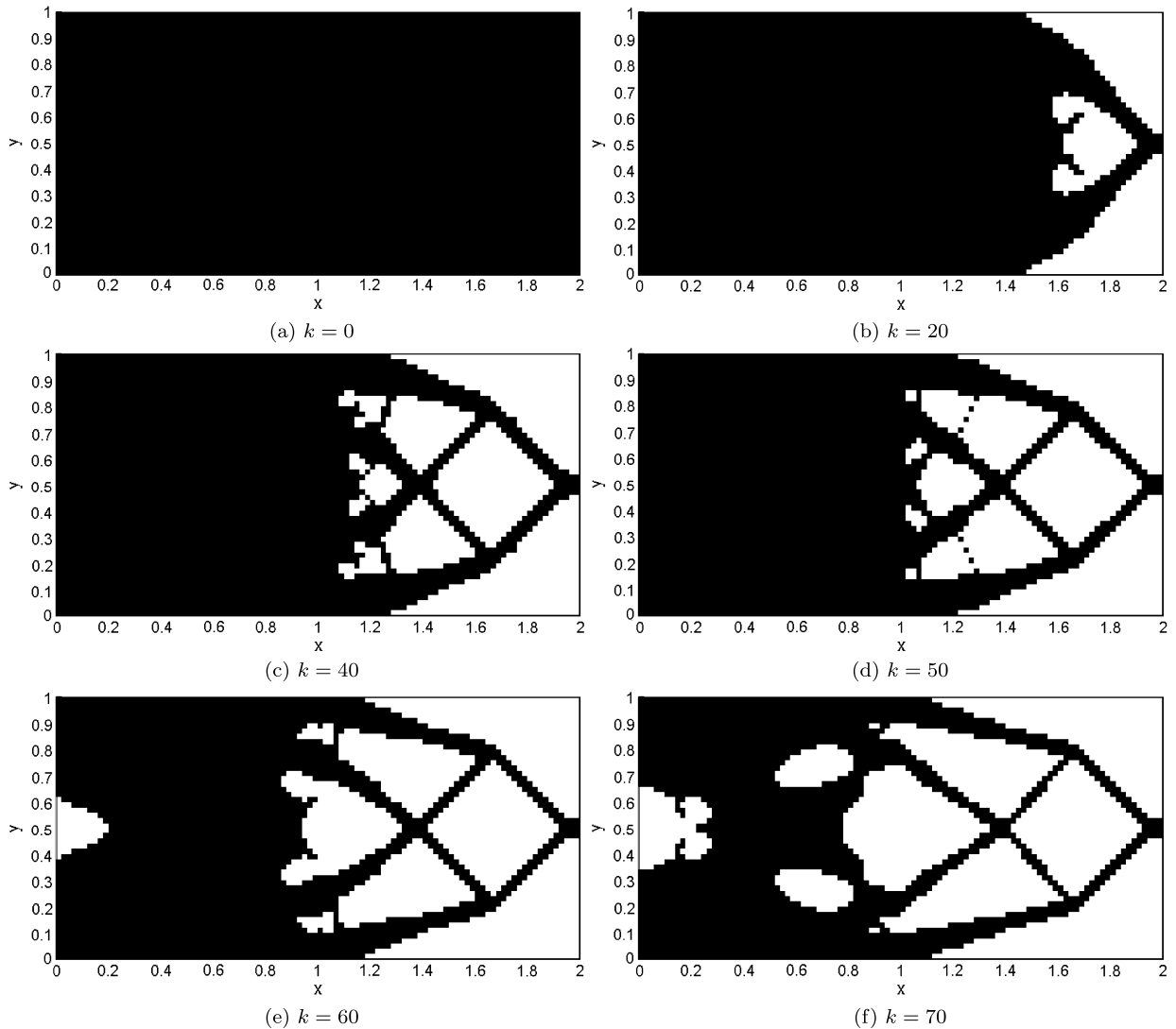
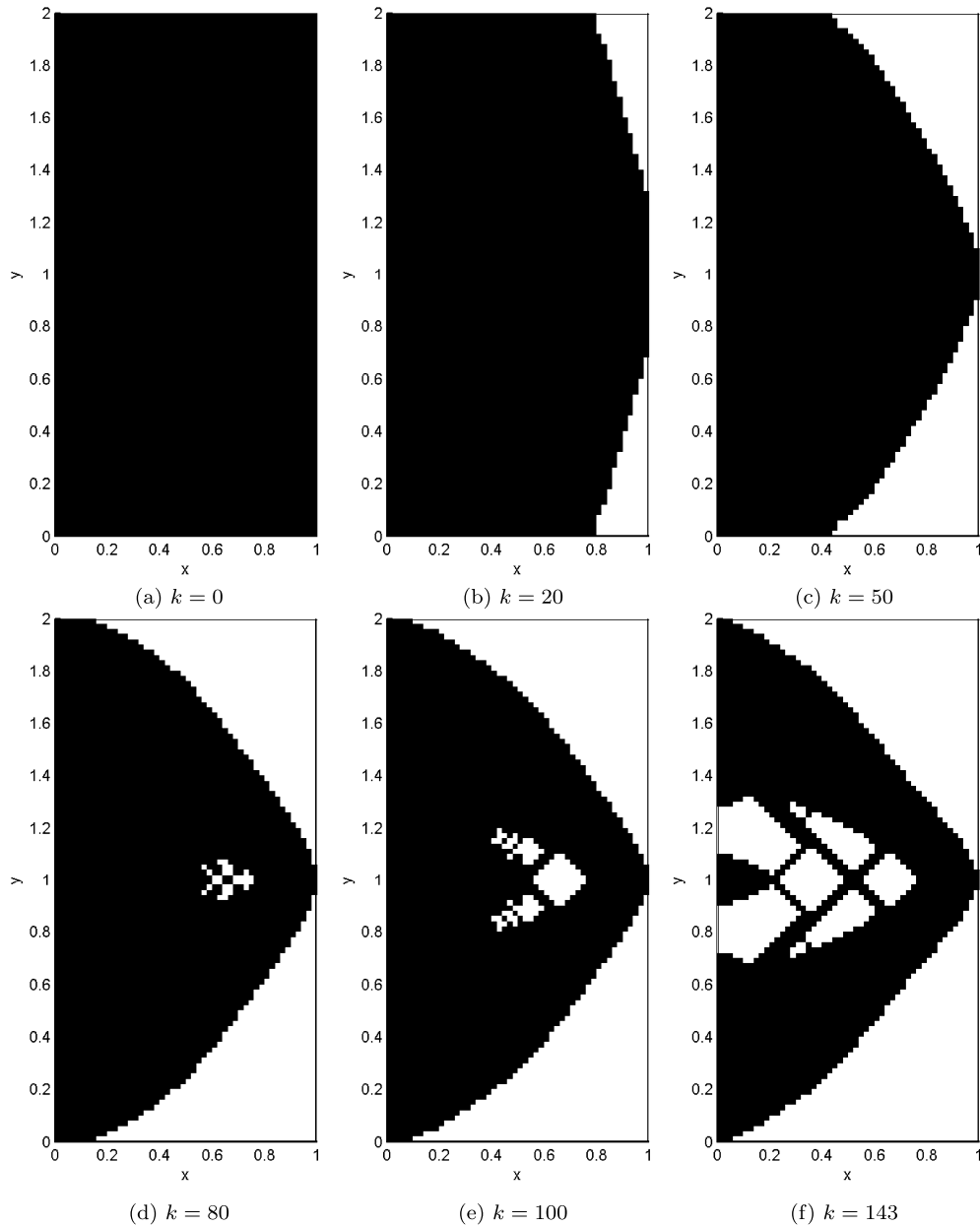


Fig. 6. The evolution of the cantilever structure  $2 \times 1$ .

eigenfrequency is  $\omega_1^2 = 1.1721E-1$ . We conduct more iterations till the 400th iteration (see Fig. 2 (f)) and observe that the extra iterations alternate the structure little.

The iteration histories of the penalty objective function and the area of original material region are shown in Fig. 3. The penalty objective function is decreasing, and the area of original material is decreasing, too. The sharp jump of the penalty objective function at around the 150th iteration is caused by the sharp area decrease of the original material region. After 194 iterations, the curves of the penalty objective function and the area of original material region become stabilized, except



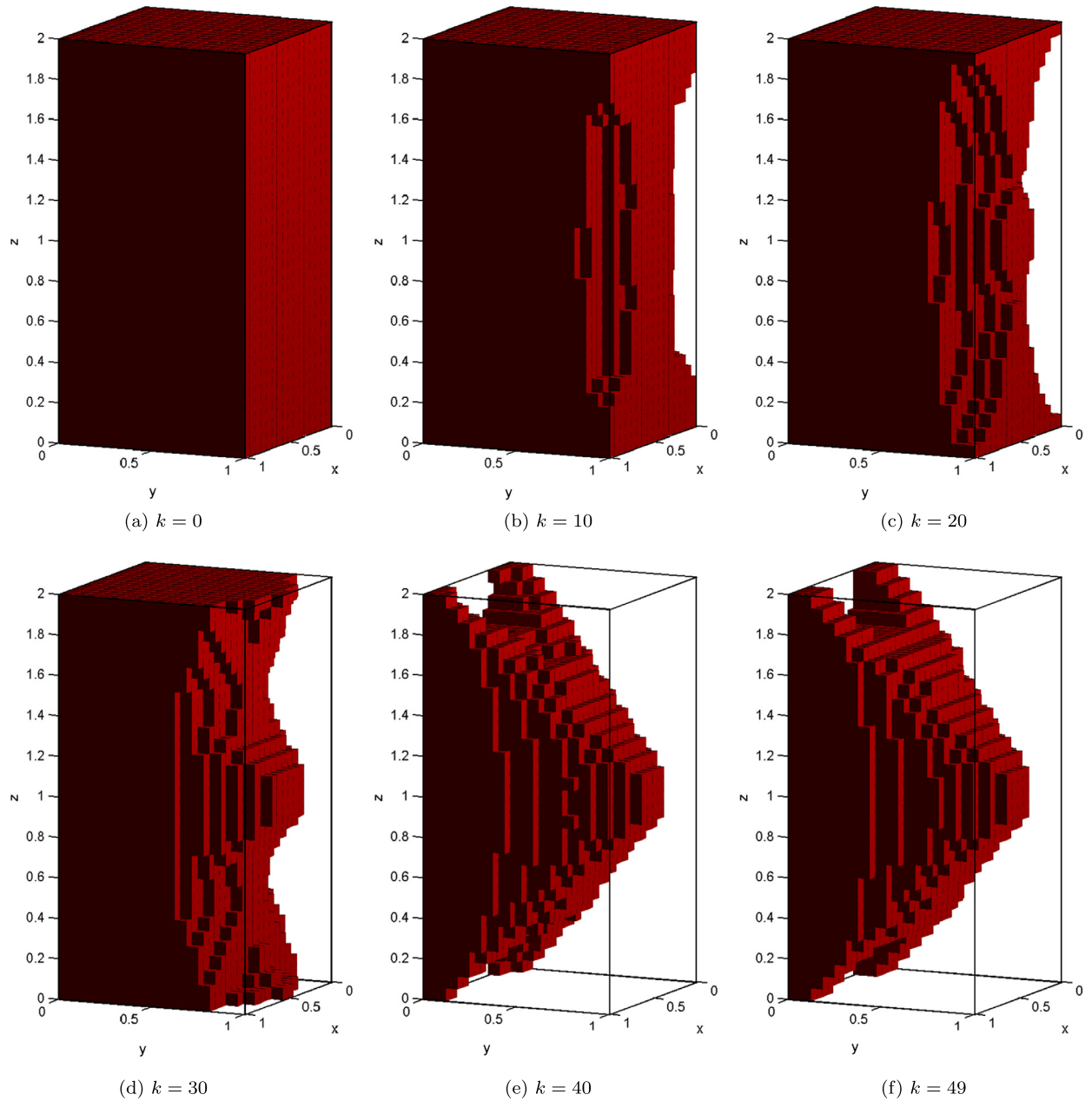
**Fig. 7.** The evolution of the cantilever structure  $1 \times 2$ .

for oscillating occasionally. Therefore, in the following numerical experiments, the stopping criterion is that the iteration is stopped if the area constraint is satisfied.

In Fig. 4, the iteration histories of the first two eigenfrequencies are illustrated. Notice that the first eigenfrequency is decreasing, not increasing. This is caused by the violation of the area constraint. That is, the area decrement of the original material may cause the first eigenfrequency decreasing, not increasing. The phenomena could also be observed in the following numerical experiments, where only the evolvments of the structure are shown.

For the second experiment, we study a medium cantilever problem. The domain is a rectangle of size  $2 \times 1$ , which is partitioned into  $100 \times 50$  rectangular elements. The left side of the rectangular domain is set to zero Dirichlet boundary condition, and the rest boundaries are set to zero Neumann boundary conditions. A small square region is set at the middle of the right side (see Fig. 5), which is 100 times heavier and is not subject to be optimized.

We set  $\alpha_{\min} = 1E-6$ ,  $\alpha_{\max} = 10$ ,  $\beta = 1E-5$ ,  $t = 6$ ,  $\zeta = 0.3$  and  $\delta = 1E-16$ . The initial PCLS function is set to  $\phi^{(0)}(x) = 1$  and is black colored in the whole background domain as illustrated in Fig. 6 (a). The evolution profiles of PCLS function are illustrated in Fig. 6 (b)–(f), after  $k = 20, 40, 50, 60$  and  $70$  iterations, respectively. The final first eigenfrequency is  $\omega_1^2 = 1.9258E-2$ .

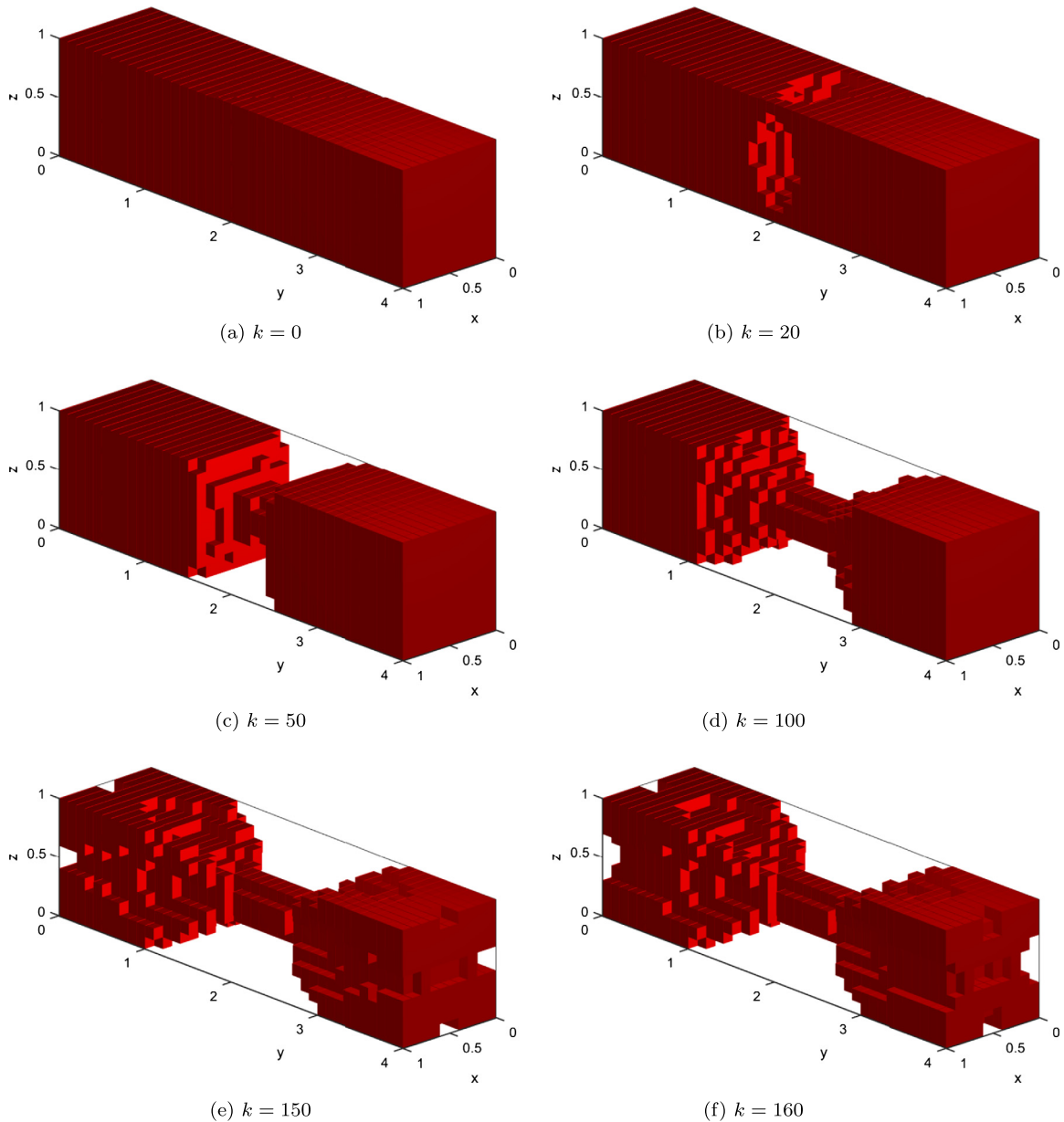


**Fig. 8.** The evolution of the 3D cantilever structure. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

For the third experiment, we run a medium cantilever problem with different size of the rectangular domain  $D$ , which is of size  $1 \times 2$ . The boundary conditions are the same as the second experiment. We partition the rectangular domain into  $50 \times 100$  rectangular elements and set  $\alpha_{\min} = 1E-6$ ,  $\alpha_{\max} = 50$ ,  $\beta = 2E-4$ ,  $t = 2$ ,  $\zeta = 0.4$  and  $\delta = 10^{-16}$ . The initial PCLS function is set to  $\phi^{(0)}(x) = 1$  and is black colored in the whole background domain as illustrated in Fig. 7 (a). The evolutions of PCLS function are illustrated in Fig. 7 (b)–(f), after  $k = 20, 50, 80, 100$  and  $143$  iterations, respectively. The final first eigenfrequency is  $\omega_1^2 = 2.8776E-1$ .

## 5.2. 3D cantilever

We optimize the first 3D cantilever. The rectangular parallelepiped domain  $D$  is of size  $1 \times 1 \times 2$ . It is partitioned into  $16 \times 16 \times 32$  cubic elements. The left side of domain is set to zero Dirichlet boundary condition, and the rest boundaries are set to zero Neumann boundary conditions. A small cubic region is set on the middle of the right side of the rectangular



**Fig. 9.** The evolution of the 3D double cantilever structure. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

parallelepiped domain (see Fig. 8), which is 100 times heavier and is not subject to optimization. We set  $\alpha_{\min} = 1E-6$ ,  $\alpha_{\max} = 100$ ,  $\beta = 5E-3$ ,  $t = 4$ ,  $\zeta = 0.2$  and  $\delta = 1E-10$ . The initial PCLS function is set to  $\phi^{(0)}(x) = 1$  and is red colored in the whole background domain as illustrated in Fig. 8 (a). The evolutions of PCLS function are illustrated in Fig. 8 (b)–(f), after  $k = 10, 20, 30, 40$  and  $49$  iterations, respectively. The red region indicates the original material, and the white one indicates the void. The final eigenfrequency is  $\omega_1^{*2} = 3.8925E-1$ .

We consider the second 3D structural optimization problem, that is, a 3D double cantilever structure. The rectangular parallelepiped domain  $D$  is of size  $1 \times 4 \times 1$ . It is partitioned into  $10 \times 40 \times 10$  cubic elements. The left and right sides of domain are set to zero Dirichlet boundary conditions, and the rest boundaries are set to zero Neumann boundary conditions. A small cubic region is set on the center of the rectangular parallelepiped domain (see Fig. 9), which is 100 times heavier and is not subject to optimization. We set  $\alpha_{\min} = 1E-8$ ,  $\alpha_{\max} = 0.3$ ,  $\beta = 1E-7$ ,  $t = 0.1$ ,  $\zeta = 0.8$  and  $\delta = 1E-10$ . The initial PCLS function is set to  $\phi^{(0)}(x) = 1$  and is red colored in the whole background domain as illustrated in Fig. 9 (a). The evolutions of PCLS function are illustrated in Fig. 9 (b)–(f), after  $k = 20, 50, 100, 150$  and  $160$  iterations, respectively. The red region indicates the original material, and the white one indicates the void. The final eigenfrequency is  $\omega_1^{*2} = 3.2026E-2$ .

## 6. Conclusion

The maximization of the smallest eigenfrequency of the linearized elasticity system is investigated. To deal with this problem, we propose a method to extend the linearized elasticity system into a fixed background domain, using a characteristic function. Different from the ersatz material method [9], the proposed method keeps the void empty, not filling the void with weak materials. Therefore, our computation is focused on the original material region, not the whole background domain. The approximation caused by ersatz material method is eliminated. Thanks to the extension, we could apply PCLS method in a fixed background domain. Then the optimization problem, which depends on the shape of the original material region, changes into an optimization problem, which depends on PCLS function. The characteristic function is represented by a quadratic function of PCLS function. The first variation of the smallest eigenfrequency with respect to PCLS function is obtained by the continuum sensitivity analysis. A penalty gradient algorithm is proposed to deal with 2D and 3D cantilever optimization problems. It is quite promising to extend our proposed method to deal with other structural optimization problems.

## Acknowledgements

This research is supported by Natural Science Foundation of Zhejiang Province, China (No. LY17A010029) and National Natural Science Foundation of China (No. 61303134).

## References

- [1] S. Zargham, T.A. Ward, R. Ramli, I.A. Badruddin, Topology optimization: a review for structural designs under vibration problems, *Struct. Multidiscip. Optim.* 53 (6) (2016) 1157–1177.
- [2] M.P. Bendsoe, N. Kikuchi, Generating optimal topologies in structural design using a homogenization method, *Comput. Methods Appl. Mech. Eng.* 71 (2) (1988) 197–224.
- [3] Y.M. Xie, G.P. Steven, Basic evolutionary structural optimization, in: *Evolutionary Structural Optimization*, Springer, 1997, pp. 12–29.
- [4] M.P. Bendsoe, Optimal shape design as a material distribution problem, *Struct. Optim.* 1 (4) (1989) 193–202.
- [5] M.Y. Wang, X. Wang, D. Guo, A level set method for structural topology optimization, *Comput. Methods Appl. Mech. Eng.* 192 (1) (2003) 227–246.
- [6] S. Osher, J.A. Sethian, Fronts propagating with curvature-dependent speed: algorithms based on Hamilton–Jacobi formulations, *J. Comput. Phys.* 79 (1) (1988) 12–49.
- [7] J.A. Sethian, A. Wiegmann, Structural boundary design via level set and immersed interface methods, *J. Comput. Phys.* 163 (2) (2000) 489–528.
- [8] S. Osher, F. Santosa, Level set methods for optimization problems involving geometry and constraints: I. Frequencies of a two-density inhomogeneous drum, *J. Comput. Phys.* 171 (1) (2001) 272–288.
- [9] G. Allaire, F. Jouve, A.-M. Toader, Structural optimization using sensitivity analysis and a level-set method, *J. Comput. Phys.* 194 (1) (2004) 363–393.
- [10] G. Allaire, F. Jouve, A level-set method for vibration and multiple loads structural optimization, *Comput. Methods Appl. Mech. Eng.* 194 (30) (2005) 3269–3290.
- [11] F. De Gournay, Velocity extension for the level-set method and multiple eigenvalues in shape optimization, *SIAM J. Control Optim.* 45 (1) (2006) 343–367.
- [12] Q. Xia, T. Shi, M.Y. Wang, A level set based shape and topology optimization method for maximizing the simple or repeated first eigenvalue of structure vibration, *Struct. Multidiscip. Optim.* 43 (4) (2011) 473–485.
- [13] L. Shu, M.Y. Wang, Z. Fang, Z. Ma, P. Wei, Level set based structural topology optimization for minimizing frequency response, *J. Sound Vib.* 330 (24) (2011) 5820–5834.
- [14] M. Theillard, L.F. Djodjom, J.-L. Vié, F. Gibou, A second-order sharp numerical method for solving the linear elasticity equations on irregular domains and adaptive grids – application to shape optimization, *J. Comput. Phys.* 233 (2013) 430–448.
- [15] Q. Xia, M.Y. Wang, T. Shi, A level set method for shape and topology optimization of both structure and support of continuum structures, *Comput. Methods Appl. Mech. Eng.* 272 (2014) 340–353.
- [16] Q. Xia, T. Shi, Optimization of structures with thin-layer functional device on its surface through a level set based multiple-type boundary method, *Comput. Methods Appl. Mech. Eng.* 311 (2016) 56–70.
- [17] M. Polajnar, F. Kosel, R. Drazumeric, Structural optimization using global stress-deviation objective function via the level-set method, *Struct. Multidiscip. Optim.* 55 (1) (2017) 91–104.
- [18] O. Christiansen, X. Tai, Fast implementation of piecewise constant level set methods, in: *Image Processing Based on Partial Differential Equations*, Springer, Berlin, Heidelberg, 2007, pp. 289–308.
- [19] J. Lie, M. Lysaker, X.-C. Tai, A variant of the level set method and applications to image segmentation, *Math. Comput.* 75 (255) (2006) 1155–1174.
- [20] P. Wei, M.Y. Wang, Piecewise constant level set method for structural topology optimization, *Int. J. Numer. Methods Eng.* 78 (4) (2009) 379–402.
- [21] X. Dai, P. Tang, X. Cheng, M. Wu, A variational binary level set method for structural topology optimization, *Commun. Comput. Phys.* 13 (05) (2013) 1292–1308.
- [22] Z. Zhang, X. Cheng, A boundary piecewise constant level set method for boundary control of eigenvalue optimization problems, *J. Comput. Phys.* 230 (2) (2011) 458–473.
- [23] A. Myslinski, Structural optimization of elastic contact problems using piecewise constant level set method, in: *CD-ROM Proceedings of the 9th World Congress on Structural and Multidisciplinary Optimization*, ISSMO, Shizuoka, Japan, 2011.
- [24] S.-H. Ha, S. Cho, Level set based topological shape optimization of geometrically nonlinear structures using unstructured mesh, *Comput. Struct.* 86 (13) (2008) 1447–1455.
- [25] S. Zhou, W. Li, Q. Li, Level-set based topology optimization for electromagnetic dipole antenna design, *J. Comput. Phys.* 229 (19) (2010) 6915–6930.
- [26] Q. Xia, T. Shi, S. Liu, M.Y. Wang, A level set solution to the stress-based structural shape and topology optimization, *Comput. Struct.* 90 (2012) 55–64.
- [27] Z. Li, T. Shi, Q. Xia, Eliminate localized eigenmodes in level set based topology optimization for the maximization of the first eigenfrequency of vibration, *Adv. Eng. Softw.* 107 (2017) 59–70.
- [28] Z. Zhang, W. Chen, X. Cheng, Sensitivity analysis and optimization of eigenmode localization in continuum systems, *Struct. Multidiscip. Optim.* 52 (2) (2015) 305–317.
- [29] S.H. Gould, *Variational Methods for Eigenvalue Problems: An Introduction to the Methods of Rayleigh, Ritz, Weinstein, and Aronszajn*, Dover Publications, New York, 1995.